Lab 1 – Requirements Engineering & UML Use-Case Modelling

#18 – Clinical Trial Subject Recruitment Engine

Use-Case Flow Specification

Use Case: Evaluate Eligibility

Primary Actor: Trial Candidate

Supporting Actor: Clinical Researcher

Supporting Use Cases: Check Inclusion Criteria, Check Exclusion Criteria

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Preconditions

1. The Trial Candidate is authorized to access the clinical trial recruitment system.
2. A valid clinical trial protocol with defined inclusion and exclusion criteria exists.
3. The Trial Candidate has provided the required survey responses for eligibility evaluation.
4. The system has access to the current trial eligibility criteria.

Postconditions

1. The candidate's eligibility status is classified successfully.
2. The applicable inclusion and exclusion criteria have been evaluated.
3. The candidate's eligibility result is stored in the recruitment system.
4. The candidate and authorized Clinical Researcher can view the resulting eligibility status.

Main Success Scenario

1. The Trial Candidate submits the required survey responses for the clinical trial.
2. The system receives and validates the submitted candidate information.
3. The system starts the eligibility evaluation process.

4. The system checks the candidate's responses against the trial's inclusion criteria.

5. The system checks the candidate's responses against the trial's exclusion criteria.

6. The system evaluates all applicable criteria.

7. The system classifies the candidate's eligibility status.

8. The system records the eligibility result and the relevant evaluation information.

9. The system displays the eligibility status to the Trial Candidate.

10. The Clinical Researcher can review the candidate's eligibility status.

11. The use case ends successfully.

Alternate Flow – Candidate Fails an Eligibility Criterion

1. During eligibility evaluation, the system identifies that the candidate does not satisfy one or more required criteria.

2. The system identifies the specific failed inclusion criterion or applicable exclusion criterion.

3. The system classifies the candidate as **Ineligible.**

4. The system records the failed criterion(s) with the eligibility result.

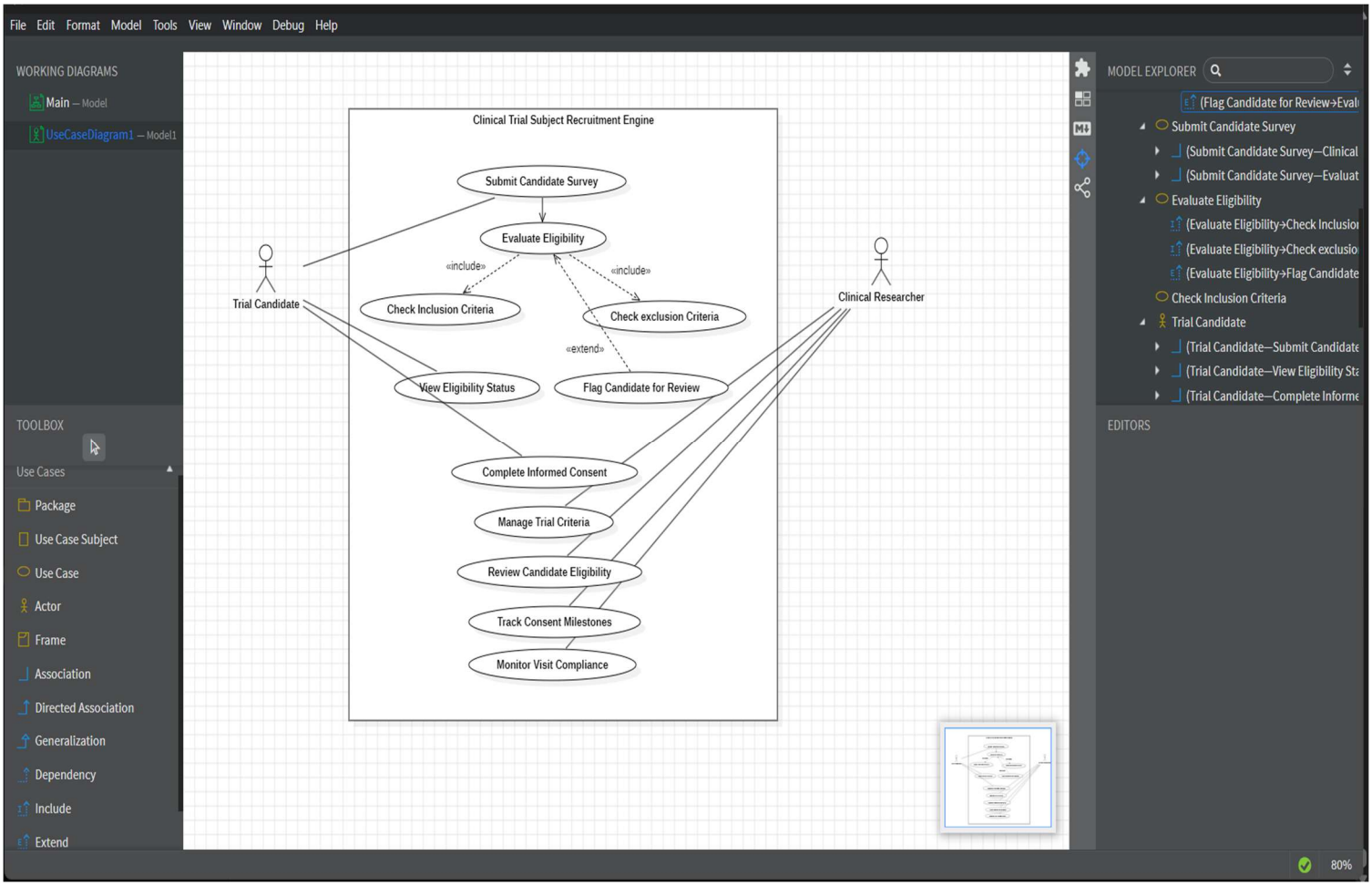
5. The system displays the ineligible status and relevant failure information.

6. The Clinical Researcher can review the candidate's eligibility result.

7. No eligible status is assigned to the candidate.

8. The alternate flow ends.

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale (short)	Comments
FR-001	Functional	The system shall evaluate candidate survey responses against trial protocol inclusion/exclusion criteria and classify eligibility status.	High	Pass: Ineligible candidate is flagged with the specific failed criteria. Fail: A subject with exclusion criteria is incorrectly marked eligible.	Establishes the core eligibility pre-screening function and prevents unsuitable candidates from being classified as eligible.	Maps to Evaluate Eligibility, Check Inclusion Criteria, and Check Exclusion Criteria.
FR-002	Functional	The system shall allow authorized users to record and track informed consent milestones for each trial candidate.	High	Pass: A consent milestone can be recorded and its current status is displayed for the candidate. Fail: An incomplete or invalid milestone is recorded incorrectly.	Ensures that informed consent progress is recorded and can be monitored throughout recruitment.	Maps to Complete Informed Consent and Track Consent Milestones.
FR-003	Functional	The system shall record and track each trial candidate's visit compliance status and associated visit logs.	High	Pass: A candidate's scheduled visit status can be recorded and retrieved correctly. Fail: A visit record is saved with missing or incorrect compliance status.	Provides a reliable record of candidate participation and helps clinical researchers monitor compliance with trial visits.	Maps to Monitor Visit Compliance.
FR-004	Functional	The system shall allow the Clinical Researcher to define and maintain the trial's inclusion and exclusion criteria used during candidate eligibility evaluation.	High	Pass: A researcher can create or update valid inclusion/exclusion criteria and the updated criteria are used during eligibility evaluation. Fail: Invalid or incomplete criteria are saved and used for eligibility classification.	Ensures that eligibility decisions are based on the current trial protocol criteria.	Maps to Manage Trial Criteria.
FR-005	Functional	The system shall allow the Clinical Researcher to review candidate eligibility status and the criteria relevant to the eligibility decision.	Medium	Pass: The researcher can retrieve a candidate's eligibility status and relevant evaluation information. Fail: The system displays an incorrect or unavailable eligibility status.	Enables clinical researchers to verify and review recruitment decisions before proceeding with trial participation.	Maps to Review Candidate Eligibility.
NFR-001	Nonfunc.	The recruitment database shall comply with 21 CFR Part 11 electronic record standards, including cryptographic audit trails.	High	Pass: Benchmarking tests confirm target latency and security standards under simulated peak load. Fail: Required security/audit checks fail or the approved latency target is exceeded under peak load.	Provides secure, traceable, and compliant electronic records for clinical-trial recruitment.	Based on the supplied NFR-001 guideline.
NFR-002	Nonfunc.	The system shall restrict access to candidate recruitment information based on authorized user roles and prevent unauthorized modification of recruitment records.	High	Pass: Unauthorized users cannot access or modify restricted candidate records, while authorized users can perform permitted operations. Fail: An unauthorized user can view or modify restricted recruitment information.	Protects sensitive recruitment information and prevents unauthorized changes to clinical-trial records.	Project-specific security requirement.


File Edit Format Model Tools View Window Debug Help
WORKING DIAGRAMS
Main — Model
UseCaseDiagram1 — Model1
Clinical Trial Subject Recruitment Engine
Submit Candidate Survey
Evaluate Eligibility
«include»
«include»
Check Inclusion Criteria
Check exclusion Criteria
«extend»
View Eligibility Status
Flag Candidate for Review
Complete Informed Consent
Manage Trial Criteria
Review Candidate Eligibility
Track Consent Milestones
Monitor Visit Compliance
Trial Candidate
Clinical Researcher
MODEL EXPLORER
(Flag Candidate for Review→Eval
Submit Candidate Survey
(Submit Candidate Survey—Clinical
(Submit Candidate Survey—Evaluat
Evaluate Eligibility
(Evaluate Eligibility→Check Inclusio
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(Evaluate Eligibility→Flag Candidate
Check Inclusion Criteria
Trial Candidate
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(Trial Candidate—View Eligibility Sta
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EDITORS
TOOLBOX
Use Cases
Package
Use Case Subject
Use Case
Actor
Frame
Association
Directed Association
Generalization
Dependency
Include
Extend
80%